

Understanding Covid-19 using data visualization

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Abstract— Coronavirus disease which is additionally called COVID-19 is a communicable disease caused by a newly discovered coronavirus. It absolutely was first identified amid a scourge of disease cases in Wuhan City, Hubei Province, China. On January 30, 2020, the WHO declared the COVID-19 outbreak a world health emergency.

The sole purpose of this dashboard is to produce researchers and enthusiasts in an area where they will analyze and visualize different aspects, trends, and patterns of covid-19.

Keywords—covid, covid-19, data visualization, interactive dashboard, dashboard

I. INTRODUCTION

Covid Dashboard is a novel coronavirus 2019 data visualization Dashboard. It is a reactive dashboard made in R using the R-Shiny framework. The sole purpose of this dashboard is to provide researchers and enthusiasts a place where they can analyze and visualize different aspects, trends, and patterns of covid-19.

Covid Dashboard is divided into five different tabs, each serving a different purpose.

A. Page1: Data Summary

This page is designed in a way to give quick information about the data of the covid-19 virus. The page will contain information like total cases, total active cases, total deaths, most affected countries, the most affected continent, world recovery date, mortality rate, etc.

B. Page2: World Data

This page is dedicated to the visualization of the world's data. This contains specially tailor-made plots for covid-19 to show all countries' situations quickly. Here users can not only see the information about any country using an informative world map but can also see the trend of the virus in different countries.

C. Page3: Visualization

This page is for detailed analysis and visualization of a specific country. Users can choose any country from the dropdown list of affected countries, and he/she can see all the relevant information with tailor-made plots. The plots will be showing daily covid data of that country. On a single plot, the user can see the regular cases, daily recovery, and daily deaths, both in cumulative and daily plots. There are other special plots to show the trend and curve of the covid-19 in that country, plot to show the curve of recovery rate, active cases, confirmed cases, and deaths.

This page also contains additional information for that country like the current recovery rate, total cases, total deaths, and active cases.

D. Page 4: Information

This page is designed to remove the misconceptions about the covid-19 virus and give user awareness about the virus and what are the safety measures to be taken in the situation of another pandemic.

E. Page 5: News

This page is to keep the user updated with the live and latest news about the virus from trusted sources. This page shows the different latest news articles.

This is an all in one dashboard for the covid-19 virus and to make sure the information is trusted, and we are not dependent on any 3rd party API. Our algorithm will automatically collect data from John Hopkins University's Github Repository (which is one of the most trusted sources for covid-19 data) and extract information from the data like each countries' daily case information, the recovery rate of each country, etc.

To further increase the user experience, all our plots will be interactive, i.e., users can engage with the plots.

II. MERITS OF PROPOSED SYSTEM

Data is the main component of our project. We need data from trusted sources to make sure information is correct, up-to-date, and easily available. [1] One of the most trusted sources for coronavirus data is from an open-

source repository maintained by Johns Hopkins University. Johns Hopkins University is a private research university in Baltimore, Maryland. Founded in 1876.

1. [2] The Data Repository is maintained by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University.
2. They daily aggregate data from different trusted sources and maintain an excel file.
3. Data is parsed into a custom made function that'll change the raw data and clean it before passing it to visualizations.
4. The Proposed system is open-source.
5. The modularity of the current system will allow us to easily add new pages and modules in future which can allow users for data analysis.

III. SOME DATA SOURCES (AGGREGATED):

- 1) [3] European Centre for Disease Prevention and Control
<https://www.ecdc.europa.eu/en/geographical-distribution-2019-ncov-cases>
- 2) [4] World Health Organization (WHO):
<https://www.who.int/>
- 3) [5] (ECDC): DXY.cn. Pneumonia. 2020.
<http://3g.dxy.cn/newh5/view/pneumonia>
- 4) [6] BNO News:
<https://bnonews.com/index.php/2020/02/the-latest-coronavirus-cases/>
- 5) [7] US CDC:
<https://www.cdc.gov/coronavirus/2019-ncov/index.html>
- 6) [8] Worldometers:
<https://www.worldometers.info/coronavirus/>
- 7) [9] India Ministry of Health and Family Welfare:
<https://www.mohfw.gov.in/>
- 8) [10] 1Point3Arcs:
<https://coronavirus.1point3acres.com/en>

After making sure we are getting proper data from a trusted source, the next challenge is to make a powerful dashboard, with user interactive graphs, data analysis, predictions, etc.

For that, we going to use R language which is one of the most used languages in data science, and for the dashboard, we are going to use Shiny which a powerful web framework for R

IV. LITERATURE REVIEW

From time to time, several notable attempts were done by various researchers for visualizing covid-19 data using various interactive dashboards. The following is a brief review of some work done for visualization of covid-19.

A. João L. D. Comba (2020) from Universidade

Federal do Rio Grande do Sul showed an example of how visualization can help understand different aspects of the pandemic.

B. [12], [13] Miguel Ribeiro (2020) from IST, University of Lisbon - ITI/LARSyS, Lisbon, Portugal done a case study on data visualization interactive exploration of human mobility data during the COVID-19 outbreak

V. METHODOLOGY

We all know about the situation the whole world faced in 2020. Coronavirus or officially called covid-19 (coronavirus disease 2019) which was first found in China in late 2019 was later become a world pandemic.

There was not a single source available which can show the data and visualization required for the analysis and predictions. When coronavirus was declared as a global pandemic, many visualization dashboards arrive within a month, but they all are lacking in many things like prediction using existing data, some are specific to a country, and those who show world data are only showing very little information like total cases, active cases and deaths.

All this information is not enough now to predict the trend. We are focused on developing a single dashboard divided into multiple sections, which will give user knowledge from the data.

VI. RESULTS

The dashboard designed in the paper was divided into 5 sections each page containing a different summary and visualization to get different levels of knowledge.

This divided styled layout helps the user to directly redirect to the level of details he's looking for. From very abstract level in first section to a very detailed visualization of the country in section 3 of the dashboard

The very first page Figure 1 which we are calling data summary is designed to show just basic information that is easy to capture at a glance and can be used for the day-to-day understanding of the covid-19 virus.

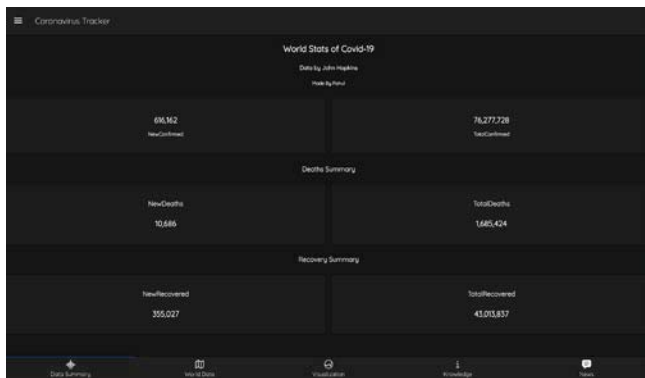


FIGURE 1. THIS FIGURE CONTAINS A SUMMARY OF COVID-19 DATA. THIS SUMMARY INCLUDES NEW CONFIRMED, TOTALLY CONFIRMED CASES, NEW DEATHS, TOTAL DEATHS, PEOPLE RECOVERED LAST DAY, AND TOTAL PEOPLE RECOVERED TILL NOW.

The second page Figure 2.1 was dedicated to showing a visualization of the overall effect of covid-19 in the world. In this section of the dashboard, you can see a world map that is colored according to the number of cases found in that country.



FIGURE 2.1. WORLD MAP SHOWING HOW THE WORLD IS AFFECTED BY COVID-19 DATA. THE COLOR IS RANGING FROM DARK BROWN (LEAST NUMBER OF CASES) TO LIGHT YELLOW (MOST NUMBER OF CASES)

The third figure in the second section of the dashboard i.e Figure 2.2 is a simple line plot showing the Cumulative trend of new cases, new recovery, and new deaths due to covid-19 throughout the world.

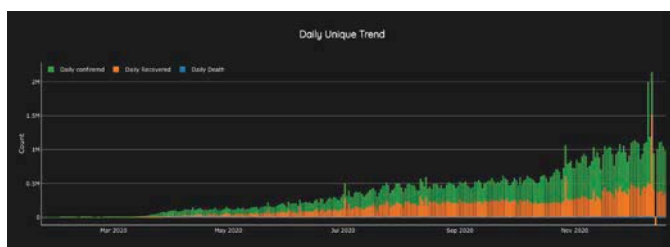


FIGURE 2.2. THIS PLOT WAS DRAWN TO SHOW THE WORLDWIDE TREND OF COVID19. ON X-AXIS WE HAVE A DATE AND ON THE Y-AXIS WE HAVE A TOTAL COUNT ON THAT DATE.

The fourth figure in the second section of the dashboard i.e Figure 2.3 is a simple line plot showing the Daily trend

of new cases, new recovery, and new deaths due to covid-19 throughout the world.

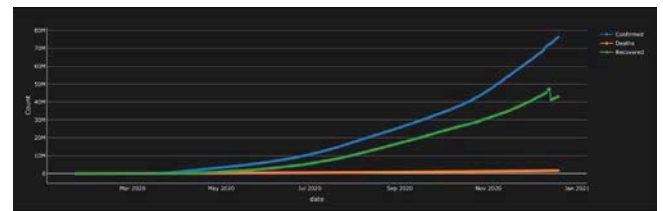


FIGURE 2.3. PLOT SHOWING THE DAILY TREND OF COVID 19

The last plot on this section Figure 2.4 was designed to show the history of top 10 affected country

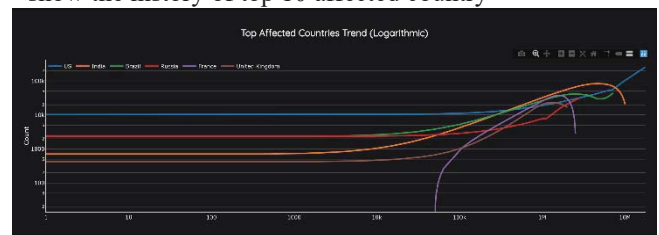


Figure 2.4. Plot showing trend of top most affected country

The third section was designed for detailed visualization of each country. Users can select different countries from the dropdown menu and can see trends of particular countries in detail.

The very first plot third section Figure 3.1 was designed to show covid data of top 10 most affected countries.

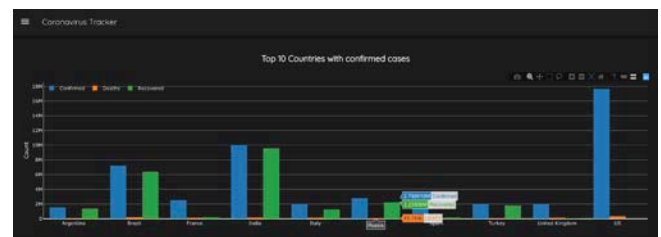


FIGURE 3.1. TOP 10 MOST AFFECTED COUNTRIES AND THEIR DATA ON BAR GRAPHS.

Figure 3.2 shows all 3 categories in bar plot.

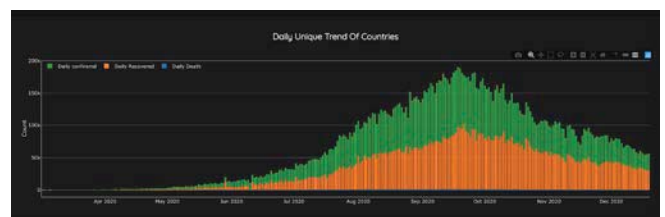


FIGURE 3.2. TREND OF COVID 19 OF SELECTED COUNTRY USING BAR PLOT.

This next image Figure 3.3 shows all 3 categories in a single cumulative line plot.

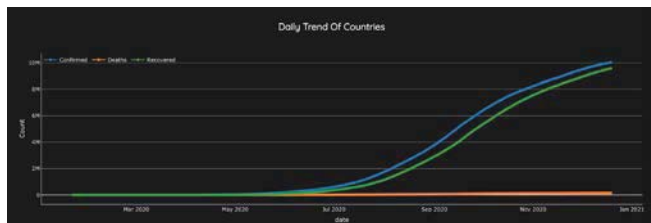


FIGURE 3.3. TREND OF COVID 19 OF SELECTED COUNTRY USING LINE PLOT.

Then the next daily trend graph Figure 3.4, is used to see the relation of daily active cases with time. As we can see, the daily cases in India have started to decrease.



FIGURE 3.4 DAILY CONFIRMED TREND OF COUNTRY

The next graph Figure 3.5 shows the trend of recovery rate in the country. We can see that the recovery rate in India has constantly increased.

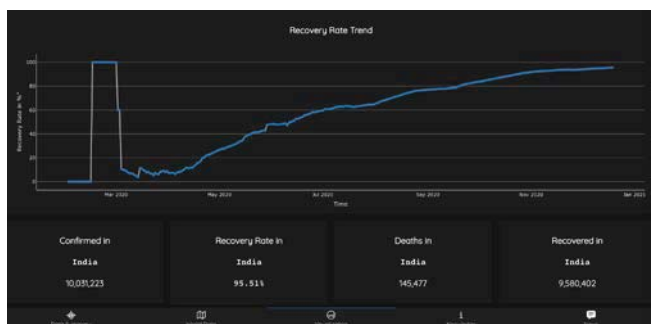


FIGURE 3.5 RECOVERY RATE TREND

The fourth section contains information and facts about covid-19. We have also included written explanations of various busted myths about covid-19.

The last section contains the latest news related to covid-19.

CONCLUSION

In this paper, firstly we demonstrated the entire number of cases of the COVID-19 together with the entire recovered and casualties across the continents with the help of plots and charts. The information presented and cases evaluated above has directed us to the inference that among various continents, Europe has been severely affected by COVID-19 pandemic with the best number of cases, and recoveries (till December 20th, 2020). Talking about countries, The USA has been severely suffering from the COVID-19 as compared to the opposite countries. various factors have led some countries to suffer more severely than others. After studying we've come to the conclusion that lockdown and travel restrictions have played a crucial role in containing COVID-19. We've also concluded that COVID-19 has the best number of cases meaning it's the best contagiousness as compared to other epidemics and has a very cheap rate.

When We visualize different countries' data, we have found that most countries stop testing covid-19. This will result in irregularity in data. Some countries like the US and India have the most accurate data. The smooth line without and outliers shows that the data tested was real and un-altered.

After visualizing different countries we can see that recovery rates vary from countries to countries. In some countries like India, the recovery rate is high as compared to population.

By observing the Daily confirmed trend plot, (fig 3.4) we can easily conclude the daily cases trend. As in fig 3.4 we can see that even when testing was constant every day, the cases started to increase from April and was on its peak during September after which it started decreasing. If we use regression to plot the future trend of active cases, we can tell that according to the current rate of decreasing in daily cases, india will start to record daily cases of less than 10,000 after march 2021.

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